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Effect of Touch Button Interface on In-Vehicle Information Systems Usability

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ABSTRACT

We analyzed the button input method of infotainment systems as part of the in-vehicle information system (IVIS) interface and related research. As IVIS becomes more complex and the need for driver-vehicle interaction increases, so does the potential risk that accompanies driver input. In this study, we propose to analyze the influence of the touch button shape, background, button and text color combination, and the presence/absence of a button border on the usability of the button input operation, which has not been studied before. Thus, participants were made to perform a button press operation in various combined conditions in a driving simulator. For keypads of the same diagonal length, we recommend a square-shaped touch keypad. In addition, the use of an edge and a specific color combination has a positive effect on the overall task performance and user satisfaction. These results are expected to be useful in the development of effective IVIS information display and input.

1. Introduction

Recently, in-vehicle information systems (IVIS), a combination of automobile and IT technologies have emerged as an increasingly important aspect of transportation technology (Horrey et al., 2006; Yoon et al., 2015a, 2015b). Cluster and head-up displays (HUD) and infotainment systems (AVN: audio, video, navigation) are commonly used in most vehicles and are commonly referred to as IVIS (Beck et al., 2019; Feng et al., 2018; Kula et al., 2017; Platten et al., 2013). A cluster is a type of instrument panel that provides detailed information about a vehicle in a driving situation (Bauerfeind et al., 2017; Broy, Alt et al., 2014; Broy, Zierer et al., 2014). An HUD utilizes augmented reality technology to display detailed information about the vehicle to the driver in the driver's forward field of view (Beck et al., 2019; Merenda et al., 2018; Schwarz & Fastenmeier, 2018; Villalobos-Zúñiga et al., 2016). An infotainment system is a device that provides entertainment functions such as in-vehicle audio, multimedia devices, and navigation devices (Gupte & Askhedkar, 2018). The goal of IVIS is to improve the driving experience through smooth interaction by displaying more information than is possible with an analogue dashboard (Feng et al., 2018; Horrey et al., 2006; Yoon et al., 2015a, 2015b).

Usability studies on the way IVIS provides information are required to ensure smooth interaction between the driver and vehicle. Cluster and HUD displays provide information visible in the driver's forward field of view (Bauerfeind et al., 2017; Beck et al., 2019; Merenda et al., 2018; Schwarz & Fastenmeier, 2018; Villalobos-Zúñiga et al., 2016). Many studies on visual complexity and visual factors have been conducted on cluster

displays (Lee et al., 2016, 2019; Yoon et al., 2015a, 2015b) and HUDs (Charissis et al., 2013; Guo et al., 2014; Merenda et al., 2018; Schwarz & Fastenmeier, 2018). Infotainment systems also have visual displays. However, additional eye movements by the driver are required to see these displays because infotainment systems are not located within the driver's forward field of view. In addition, complex IVISs provide various types of information, such as entertainment information, in addition to information directly related to driving, and these types of systems involve manual control using physical buttons. Recently, several studies have been conducted on the visual area of touch screen-type infotainment systems (Graichen et al., 2019; Lee & Ji, 2019). However, unlike clusters and HUDs, infotainment systems have various functions, so physical control by task is required (Mathur et al., 2017). In most vehicles, the infotainment system is operated by means of a physical keypad. However, the importance of physical buttons is decreasing as efforts continue to minimize drivers' movements and reduce workload (Holzinger, 2002; Kim et al., 2014; Scott & Conzola, 1997). Increasingly, touch screens are replacing physical buttons (Suh & Ferris, 2019). For this reason, many functions are introduced to the infotainment system of the center fascia. Not only numbers but also various character inputs are required. However, unlike physical buttons that provide both visual and nonvisual sensations, buttons on a flat touch screen require more visual attention from the driver, exposing the driver to safety problems (Fitch et al., 2013; Suh & Ferris, 2019). In addition, a number of studies show that touch buttons do not perform as well as physical buttons (Jung & Im, 2015; Suh & Ferris, 2019; Tao et al., 2018).

Therefore, research on IVIS interaction to improve work performance should be conducted to improve driving safety.

A concern regarding IVIS is that it tends to reduce driving safety by increasing the eye-off-the-road time (EoRT) (National Highway Traffic Safety Administration, 2012). Therefore, most IVIS studies focus on ensuring driving performance and safety. Some principles have been established concerning how best to maintain driving safety and driving performance. According to the Society of Automotive Engineering (SAE), to ensure driving safety, it should be possible to perform tasks on vehicle displays and manual control equipment in less than 15 s (Society of Automotive Engineers, 2004). The National Highway Traffic Safety Administration (NHTSA) has set limits for glance duration time (less than 2 s on average), Number of glance time (less than 15% of the total), and total eyes-off-road time (TEORT; 12 s) for in-vehicle devices (National Highway Traffic Safety Administration, 2012). A number of IVIS studies have also attempted to provide guidance by examining driving performance and risks in relation to touch button size (Feng et al., 2018; Kim et al., 2014) and layout (Li, Chen et al., 2017). As more and more functions are added to the actual vehicle's IVIS, combinations of numerous factors (i.e., button size, shape, color, border, symbol, etc.) are used in vehicle interfaces. Accordingly, ineffective combinations can affect the driver's EoRT, and ultimately, driving safety. However, existing studies have been conducted on only a few factors; hence, the results are limited and of restricted applicability. Therefore, to provide a criterion for an interface design element rather than a performance criterion such as a task completion time or a glance duration time, it is necessary to assess comprehensively the influences of various factors related to performance, including correlations among factors. Against this background, we attempt to understand the influence of various design factors of the touch button interface of the infotainment system on usability, and conduct a study on IVIS interactions that can represent even character input through the numeric keypad.

2. Literature review

A number of interaction studies have been conducted on infotainment systems. Various infotainment systems have been studied, such as touch buttons (Crundall et al., 2016; Feng et al., 2018; Kim et al., 2014; Suh & Ferris, 2019), overall user interface (UI) elements (Hua & Ng, 2010; Naujoks et al., 2019; Pankok & Kaber, 2018), icons/symbols (Silvennoinen et al., 2017), screen position (Kuiper et al., 2018), layout (Kim et al., 2015; Li, Chen et al., 2017), sound effects (Larsson & Niemand, 2015), gestures

(Graichen et al., 2019; Parada-Loira et al., 2014), and modality (Gaffar & Kouchak, 2017).

Research on touch buttons has been conducted in a number of ways in terms of size factors, using Fitts' law (Crundall et al., 2016; Feng et al., 2018; Kim et al., 2014; Suh & Ferris, 2019). Touch button size has been found to be a major factor in research on kiosks (Colle & Hiszem, 2004) and mobile phones (Kim et al., 2018). A series of studies related to IVIS touch buttons have also been conducted. Kim et al. (2014) conducted a usability evaluation that considered three speed conditions (0, 50, and 100 km/h) and five square touch button sizes (7.5, 12.5, 17.5, 22.5, and 27.5 mm). Based on the results, a touch button size of 17.5 mm or more was recommended. A 17-inch computer monitor was used as the screen type, and the task was a 5-key input. Feng et al. (2018) conducted an IVIS usability evaluation of three square touch button sizes (14, 24, and 33 mm) and three touch button counts (4, 8, and 15). Based on the results, a touch button size of more than 24 mm and 4 to 6 touch buttons were recommended. An 8-inch Smartphone was used as the screen type, and the task was button search and press. Suh and Ferris (2019) evaluated IVIS usability for two square touch button sizes (15 and 25 mm) and found that, in terms of performance, a touch button size of 25 mm was generally more usable, but that touch button size made no difference when sound effects were added. Smartphone and physical keypad were used as screen type, and the task was the 5-key input method. Crundall et al. (2016) evaluated IVIS usability for five touch screen font sizes (4, 5, 6.5, 8, and 9 mm), and based on the results, recommended a font size of 8 mm or more. A 10.1-inch Smartphone was used as the screen type, and the task was text reading. However, there are differences among the recommendations derived from these different studies because of differences in the experimental environments and tasks assessed (Table 1).

For touch buttons, the NHTSA recommends an optimal visual angle of 86 arcmin and a minimum visual angle of 41 arcmin (Campbell et al., 2007). However, to apply this guidance to an actual vehicle system, it is necessary to consider the influence of various factors other than size. The studies related to touch buttons summarized above also tried to confirm associations among factors by examining specific situations. For example, Kim et al. (2014) conducted a cross-study of button size and vehicle speed to consider the driving situation. However, although the effect of the size of the touch button was determined, no significance was detected for the speed as an independent variable. Feng et al. (2018) considered the number of touch buttons as an independent variable, but the number of buttons is limited

Table 1. Research on touch button size in IVIS.

Focus	Factor and their levels	Recommendations	Screen Type	Task Type	Author
Touch button	Size: 7.5, 12.5, 17.5, 22.5, 27.5 (mm)Speed: 0, 50, 100 (km/h)	17.5 mm [†]	Computer Monitor (17 inch)	5-key Input	Kim et al., 2014
	Size: 14, 24, 33 (mm)Button quantity: 4, 8, 15	24 mm [†] Quantity = 4–6	Smartphone (8 inch)	Button Search, Press	Feng et al., 2018
	Size: 15, 25 (mm)Type: Touchscreen, physical keypad	15 < 25 mm (general)15 = 25 mm (+ sound effect)Touchscreen < physical keypad	Smartphone (-)Physical Keypad (-)	5-key Input	Suh & Ferris, 2019
Font	Size: 4, 5, 6.5, 8, 9 (mm)	8 mm [†]	Smartphone (10.1 inch)	Text Reading	Crundall et al., 2016

as a variable in that it has low effectiveness for functions such as a fixed-number dial. Suh and Ferris (2019) presented a cross-study of button size with sound effects but detected no significant difference among button sizes when sound effects were used.

In addition to size in interface design elements, attention to color has been made in various studies (Akçay et al., 2012; Chi & Dewi, 2014). Color-related characteristics have a significant effect on the usability of interfaces, including automobiles. Generally, if many colors are used in IVIS, it can be more easily distinguished (Konrad & Bruce, 2001); however, excessive use can cause confusion (Park & Park, 2019). In addition, different results may be presented depending on the color collision (contrast), intensity, chrominance, etc. (Berns, 2019; Jonauskaitė et al., 2019). Color plays an important role in a product, accounting for more than 60% of the user's purchasing decision factor (Singh, 2006); among various design elements, color accounts for a very high proportion (Akçay et al., 2012). From the perspective of the IVIS interface, research on the use of color in cluster displays and HUDs has been conducted (Table 2). Smith and Fu (2011) evaluated preferences for three single colors (orange, blue, and green) for the color of the cluster's speedometer. In terms of modern sensibility, orange and green were recommended. Li, Qu et al. (2017) recommended black and gray colors for the background of the dashboard, based on a qualitative analysis of dashboard colors for a cluster display. Huang et al. (2013) conducted a preference evaluation and response time evaluation for four single colors (orange, green, blue, and white) for an HUD speedometer. Green yielded the shortest response time, but white was the best in terms of safety. The IVIS interface used in an actual vehicle involves a combination of a background, buttons (icons or symbols), and text colors. However, the studies mentioned above considered only a single color and not the effect of color contrast.

Another interesting issue is the IVIS usability research methodology used. In most IVIS usability studies, quantitative and qualitative data are collected simultaneously to interpret the results. For quantitative data measurement and collection, a driving simulator and eye tracker equipment are mostly used (Broy, Alt et al., 2014; Kim et al., 2015; Larsson & Niemand, 2015; Lee et al., 2019; Li, Chen et al., 2017; Pankok & Kaber, 2018). Although studies using actual vehicles have been conducted (Kuiper et al., 2018; Mourad et al., 2017), quantitative data collection using a driving simulator and eye tracker is more common. Task completion time (Angelini et al., 2016; Jeong & Liu, 2017; Lee et al., 2019; Li, Chen et al., 2017; Weidner & Broll, 2018) and error rate (Angelini et al., 2016; Kula et al., 2017; Lee et al., 2019; Li, Chen et al., 2017; Weidner & Broll, 2018) are commonly used as dependent variables in assessing performance in IVIS tasks. The IVIS touch button studies closely related to the research described in this paper have mainly used task completion time as a dependent variable (Crundall et al., 2016; Feng et al., 2018; Kim et al., 2014).

In addition to task performance, most IVIS studies have used a separate measure for subjective evaluation of the driver's workload. Table 3 summarizes measures used in workload evaluation. The NASA-TLX (Task Load Index) developed by the National Aeronautics and Space Administration (NASA) is the most commonly used measure of a driver's workload (Jakus et al., 2015; Kim et al., 2014; Lee et al., 2019; Pampel et al., 2019; Villalobos-Zúñiga et al., 2016). NASA-TLX is a subjective workload evaluation tool that characterizes perceived workload based on responses to six questions concerning mental demand, physical demand, temporal demand, performance, effort, and frustration. Although a number of studies have been conducted using NASA-TLX, the reduced clutter score (RCS) questionnaire is also used in IVIS to assess display clutter (Pankok & Kaber, 2018). RCS is an evaluation tool that measures driver perception

Table 2. Research on color in IVIS.

Device	Element	Recommendations	Reference
Cluster	Dashboard color	Background color: black, gray	Li, Qu, et al., 2017
HUD	Speedometer color: orange, green, blue	Orange, green	Smith & Fu, 2011
HUD	Speedometer color: green, orange, blue, white	Green, white	Huang et al., 2013

Table 3. Cognitive load measurement tools.

Questionnaire	Measure	Description	Author
NASA-TLX	Mental Demand	What is the level of mental and cognitive burden?	Hart & Staveland, 1988
	Physical Load	How much physical activity is required?	
	Temporal Demand	The extent to which time pressure is felt	
	Effort	The level of effort made to achieve a successful task	
	Performance	How successful the outcome of the work has been to achieve the goal of the work	
RCS	Frustration	Comprehensive evaluation of anxiety, frustration, and anger during work	Kaber et al., 2013
	Overall Clutter	How information is displayed is distracting and complex	
	Dynamics/Variability	How often the information display method changes	
	Consistency/Similarity	How inconsistent is the way information is displayed?	
INTUI Subjective Rating	Colorfulness	How many different colors are used to display information?	Ullrich & Diefenbach, 2010 Chi & Dewi, 2014; Francois et al., 2017
	Intuition	Whether the vehicle interface is intuitive and easy to understand	
	Familiarity	Whether the vehicle interface is familiar with performing tasks	
	Learnability	Whether it is easy to learn to familiarize yourself with the vehicle's interface	
	Overall Usability	Whether the overall use of the vehicle interface is comfortable	

due to momentary confusion (Kaber et al., 2013; Pankok & Kaber, 2018) based on the response to four questions concerning overall clutter, dynamics/variability, consistency/similarity, and colorfulness.

Subjective questionnaires are used to measure user experience in driving situations. The intuitive interaction (INTUI) questionnaire is a tool used to measure the intuitiveness of the UI interaction (Ullrich & Diefenbach, 2010). INTUI is also used in vehicle IVIS evaluation (Broy, Alt et al., 2014). In addition, familiarity, learnability, and overall usability are used as individual subjective evaluation scales in vehicle interface evaluation (Chi & Dewi, 2014; Francois et al., 2017).

3. Methods

3.1. Participants

Twenty Korean university students participated as participants in the experiment. The gender ratio of the participants was uniform. The average age was 22.75 yr (standard deviation ± 1.89 yr). Every participant had a legal driving license, at least 1 yr of driving experience, and no visual impairments. The experiment took approximately 2 h for each participant to complete, including break time.

3.2. Apparatus

A driving simulator, eye tracker, and smartphone device were used for the experiment (Figure 1). The driving simulator (Gridspace Inc., Model Sedan-2014-D) consisted of a Hyundai Sonata's driver seat (NF Sonata) and three 42" light-emitting diode (LED) monitors. The simulator's software produces high-quality video (1920×1080 pixels) using a three-dimensional (3-D) graphics engine as well as various sounds (e.g., engine noise, horn sounds, a tire friction noise during sudden starting or stopping). The driving scenario was built into the driving simulator and consisted of Driving at straight road, stop at the traffic light, and Driving at roundabout. The road width of the simulator is approximately 3.5 m, which is the same as the actual width of a Korean highway traffic lane. Through the left and right monitors, it is possible to use a rearview mirror, as in a real vehicle, and to



Figure 1. Experimental environment.

change lanes smoothly. The speed and engine revolutions per minute (RPM) can be checked visually through a linked cluster display. Using Tobii Pro Glasses 2 as the eye tracker, user visual change data were collected through one-point calibration. Each participant participated in the experiment wearing an eye tracker under the control of the experiment performer. The software of the IVIS prototype was created as an application using Android Studio, running in Android OS version 28 on a Samsung Galaxy Note 8 (SM-N950N; screen size 6.3 in; resolution 2940×1440). The IVIS prototype was firmly fixed to the top of the center fascia.

3.3. Experimental design

The factors considered touch button shape (hereinafter referred to as Shape; two levels), edge presence or absence (hereinafter, Edge; two levels), and color combination (hereinafter, Color; five levels), for 20 factor level combinations. The application prototype was designed with the vertical phone dial motif commonly used in vehicle IVISs. Only 3×3 keypads with numbers from 1 to 9 were used; 0, *, and # were not used in the experiment. The design details for the three factors are as follows.

3.3.1. Touch button shape and font design

The universal rectangle and square shapes were selected as the touch button shapes. The button size was selected with reference to the touch button visual angle standard set by NHTSA (Campbell et al., 2007). To apply this standard to the equipment used in this study, the visual angle was converted to a button size in the following manner. First, the average distance from the eye level to the IVIS prototype at the top of the center fascia for five people (average age = 23.4; standard deviation ± 1.35) in the same age group as the participants of this study was determined as 75 cm. The recommended and minimum button sizes (18.7 and 9 mm, respectively) were derived accordingly. Based on the size of the touch button, a length for each side, depending on the shape (rectangular or square), was selected. The total diagonal length of the rectangular and square 3×3 keypads had to be the same so that a participant's finger travel distance would be the same for both. Therefore, the design for a rectangle touch button was developed first, and the design for a square touch button was then developed by combining the recommended horizontal length in the case of a rectangle and the minimum vertical length (recommended size 18.7 mm, minimum size 9 mm). The buttons were spaced 6 mm apart (Jin et al., 2007). Based on the design for a rectangular button, a 3×3 keypad was designed. The width and length of the keypad were 68.1 mm and 39 mm, respectively, and the diagonal length was 78.5 mm. A 3×3 square keypad was designed using a diagonal length of 78.5 mm, and the length of one side of a square button was selected to be 14.5 mm.

The font size for the touch button was determined with reference to the font standard prescribed by the NHTSA (Campbell et al., 2007; recommended visual angle 20; minimum visual angle 12 arcmin) in the same manner as the touch button size. The font size was derived using the average

distance from the eye height to the prototype (75 cm), which resulted in a recommended font size of 4.3 mm, and a minimum font size of 2.6 mm. The former, 4.3 mm, was selected as the font size, considering the button size. San Serif was used as the font type (Naujoks et al., 2019).

3.3.2. Touch button edge design

To assess the effect of the presence or absence of an edge, designs with and without edges were developed for the rectangular and square button shapes. To prevent affecting the size of the button, the inner edge of the button was marked with an edge. The thickness of the edge was fixed at 1 mm.

3.3.3. Color combination design

Objective indicators were employed to select a color combination to be used as the foreground and background of the text. Before determining the contrast ratio values for the two colors, the colors to be used were selected. First, ten chromatic colors (red, orange, yellow, yellow-green, green, cyan, blue, indigo, purple, and wine), and three achromatic colors (white, gray, and black) were selected based on the IRI Hue & Tone 120 system. As the primary color has a brightness value of 100%, there is a problem in using it, so a standard for a representative color to replace it with was selected. In this study, the so-called VIVID color set was selected as corresponding to the highest saturation (sharpness) with the brightness reduced as much as possible, considering the driving environment. After confirming the red-green-blue (RGB) values for 13 VIVID colors (ten chromatic colors and three achromatic colors), a new contrast ratio table was constructed by calculating the contrast ratio according to the CIE 1976 standard (Table 4). Based on research results that recommend

color contrast ratio of 5 or more for night driving, combinations of colors with contrast ratios exceeding this value were used in the study (Naujoks et al., 2019).

This study investigated the general color combinations used in the infotainment systems of vehicles on sale. Universal touch button colors used for the background color, font color, and edge color for the latest releases of nine automobile models (BMW 730d, Benz S450, Tesla S100D, Hyundai Sonata DN8, Land Rover Discovery 5 TD6, Porsche Cayenne, Audi A8, Volvo XC60, and Honda Accord) were determined. Interfaces such as alerts and warnings for specific situations were not considered. Black (contrast ratio of 8) was most commonly used for the background color. Gray (contrast ratio of 5) and blue (contrast ratio of 4) were most commonly used for the buttons. White (contrast ratio of 8) was most commonly used for the font. White (contrast ratio of 8) was most commonly used for the edge, and orange (contrast ratio of 3) was also used.

Five color combinations were selected using the contrast ratio table and the results of the vehicle survey (Table 5). Red and yellow, with very low preference and sensitivity scores, were excluded regardless of the contrast ratio (Tsang et al., 2013). In the case of the edge color, only two levels exist, so the second universal index, orange, was used based on consideration of the whiteness and overlap of the font. Combinations 1 and 3 were selected as controls to compare the effect of 'white, black', which has a contrast ratio of more than 5. Gray, the most common button color, was used for the button color for this combination. Combination 2 was selected to confirm the effect of 'purple, white', with a contrast ratio of more than 5. Purple was used for the button color because there is no actual vehicle

Table 4. Contrast ratio table.

Color(VIVID)	Red	Orange	Yellow	Yellow-Green	Green	Cyan	Blue	Indigo	Purple	Wine	White	Gray	Black
Red	-	2.46:1	4.05:1	2.77:1	1.63:1	1.28:1	1.88:1	1.16:1	1.03:1	1.2:1	a5.78:1	2.24	1.53:1
Orange		-	1.64:1	1.12:1	1.5:1	1.91:1	1.3:1	2.11:1	2.54:1	2.04:1	2.34:1	1.1:1	3.78:1
Yellow			-	1.46:1	2.47:1	3.15:1	2.15:1	3.47:1	4.18:1	3.36:1	1.42:1	1.8:1	a6.21:1
Yellow-Green				-	1.69:1	2.15:1	1.47:1	2.37:1	2.86:1	2.3:1	2.08:1	1.23:1	4.25:1
Green					-	1.27:1	1.15:1	1.4:1	1.69:1	1.36:1	3.53:1	1.36:1	2.51:1
Cyan						-	1.46:1	1.1:1	1.32:1	1.06:1	4.49:1	1.74:1	1.97:1
Blue							-	1.61:1	1.94:1	1.56:1	3.06:1	1.18:1	2.89:1
Indigo								-	1.2:1	1.03:1	4.95:1	1.91:1	1.78:1
Purple									-	1.24:1	a5.96:1	2.31:1	1.48:1
Wine										-	4.8:1	1.86:1	1.84:1
White											-	2.58:1	a8.87:1
Gray												-	3.43:1
Black													-

aContrast ratio greater than 5

Table 5. Final color combination level.

Combination	Background	Button	Font	Edge	Benchmark	Abbreviation
1	White	Gray	Black	Orange	Contrast: Best Actual Car: Common	WGB
2	Black	Purple	White		Contrast: Best	BPW
3	Black	Gray	White		Contrast: Best Actual Car: Common	BGW
4	Black	Blue	White		Actual Car: Common	BBW
5	Black	Yellow-Green	White		Contrast: Worst Actual Car: Common	BYW

case where purple was used as the font system. The most common background color, black, was used for this combination. Combination 4 was selected to confirm the effect of blue, which is used as the actual button color of the vehicle. For comparison according to contrast ratio, yellow-green was selected as Combination 5. White, the most common font color, was used as the font color for this combination.

3.3.4. Final application design

The touch button interface was designed with the 20 combinations of the levels of the three factors considered (Figure 2). Figure 2(a) illustrates each part of the keypad, and (b) shows the 20 design alternatives.

3.3.5. Independent and dependent variables

The independent variables were the touch button shape (rectangle, square; two levels), color combination (Table 5; five levels), and edge presence or absence (two levels). The dependent variables were the task completion time per button, the number of errors per button, eye-tracking data, and scores for each questionnaire item. For each button operation, the pressed button and the time the button was pressed were automatically saved. The mean task completion time and error count data for each button were determined. This study focused on the first task among the buttons. For the mean completion time, the completion time for the first button and the completion time for all buttons were classified and used for analysis. The error rate was calculated by comparing the five-digit

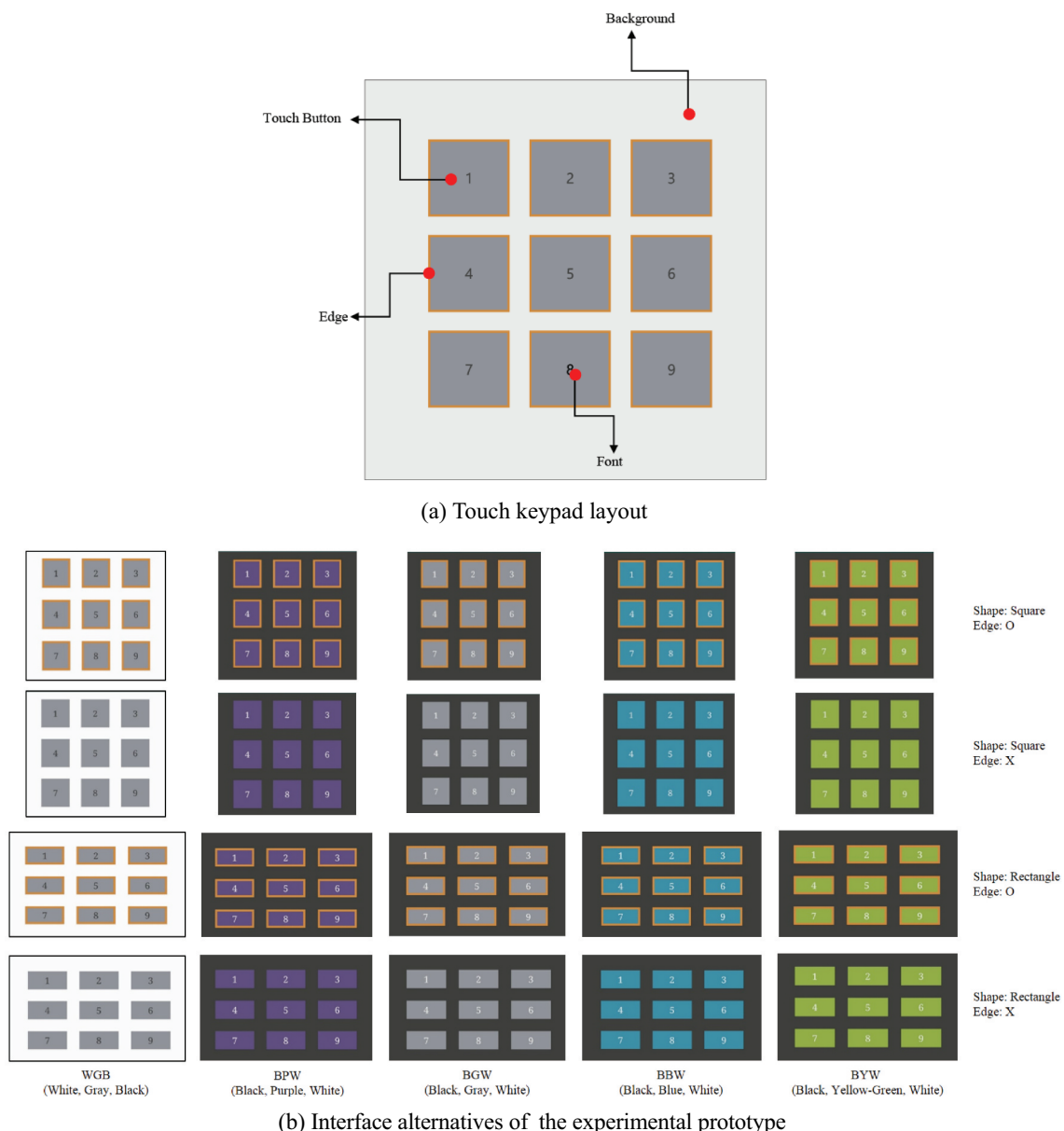


Figure 2. IVIS prototypes.

number presented to the participant with the actual input string. Duplicates, omissions, or other wrong button inputs were considered errors. The eye tracking data collected were the mean glance time and number of glances for on-road and off-road conditions (Table 6).

NASA-TLX, RCS, and DX questionnaires were used for the survey items. A scale of 0–100 was used for evaluation for each combination of independent variables. Six NASA-TLX questions concerning mental demand, physical demand, temporal demand, effort, performance (degree of failure), frustration, and overall load were used to measure the cognitive load. To evaluate the clutter of the vehicle display, four RCS questions concerning overall clutter, dynamics, inconsistency, and colorfulness were used. To measure the user experience when driving, the items of INTUI were reorganized and named as DX (driving experience) questionnaire. The DX questionnaire consisted of four questions concerning familiarity, learnability, intuition, and overall usability (Table 6).

3.4. Experimental procedures and tasks

There were three preliminary stages of the experiment. First, the participant was instructed to write down personal information (driving experience and other personal information) and then select five numbers to be used for the touch button operation. The participants were asked to enter any five digits they wanted, except that they were restricted from using simple patterns (such as 12345, 13579, or 11111). The five selected numbers were memorized by the participant and used in the same way in subsequent repeated experiments. Second, detailed explanations were provided to the participants on the purpose, procedures, and precautions of the experiment. The precautions included 1) compliance with road regulations, 2) mandatory use of gears and turn signals, and 3) prohibition of excessive speeding. Third, the experiment was conducted after providing sufficient explanation and practice time to exclude the influence of the difficulty of the driving method and the scenario used in the driving simulator.

Table 6. Taxonomy of whole measures.

Data classification	Indicator
Task performance(First button presses)	Mean completion time (s) Error rate (%)
Task performance(All button presses)	Mean completion time (s) Error rate (%)
NASA-TLX(0–100 Score)	Mental demand Physical demand Temporal demand Effort Performance (degree of failure) Frustration
RCS(0–100 Score)	Overall clutter Dynamics Inconsistency Colorfulness
DX(0–100 Score)	Familiarity Learnability Intuition Overall usability
Eye tracking data	On-road mean glance time (s) Number of on-road glances Off-road mean glance time (s) Number of off-road glances

In this preliminary phase, before the driving scenario was started, one-point calibration was performed to correct different gaze patterns by adjusting the eye tracker's origin. Subsequently, while driving according to the virtual scenario, the participant performed a touch button pressing operation. The map and work composition for the driving scenario are as follows (Figure 3). The driving road consists of two straight sections and one curved section in between. There is a traffic light between the straight section and the curved section to make the driver pause for a while. For the sake of immersion in the driving experience, the entire route included in the scenario was driven, but considering the effect of difficulty, the touch button operation was performed only while driving in a straight line. In each straight driving section, an experiment was conducted for one condition. Because the driving scenario included two straight sections, two conditions could be evaluated for each scenario. To carry out 20 conditions, 10 identical driving scenarios were repeated. The same button press operation was repeated twice within each condition (Figure 4). The length of the straight line was sufficient for the participant to perform the touch button operation twice (The average time for all driving runs completed was 180 s; the average time for driving in a straight line was 30 s.). To control the effect of color on eye and task fatigue, a break of 2 min was given for each scenario. As there are many items in the questionnaire, a participant's concentration may be diminished, so the experiment performer explained each question in detail and asked for an accurate response. In addition, one-point calibration was repeated for each scenario for accurate measurement. The progression of all conditions excluded the effect of the sequence by using a Latin square.

4. Results

4.1. Task performance

A three-way repeated-measures ANOVA was conducted to check whether different combinations of the three independent variables (Shape, Edge, and Color) produced significantly different values for two of the dependent variables (task completion time per button and number of errors) (Table 7). A post hoc Student Newman–Keuls (SNK) analysis was also conducted. When an interaction was significant, a simple effect analysis was also performed. Since the visual and physical requirements for pressing the first button are different from pressing the next button, the time taken to press the first button and the total task completion time were analyzed separately. The analysis of the mean completion time of first button presses showed that Edge, Color, and Edge $\times$ Color were statistically significant. When a button edge was present, the mean completion time was shortened ($F = 5.069$, p -value < 0.05). When the color combination was 'background-color: black, button color: blue, text color: white' (BBW), the mean completion time was greater than that for other color combinations ($F = 2.576$, p -value < 0.05). A simple effect analysis of Edge $\times$ Color showed that it was not statistically significant

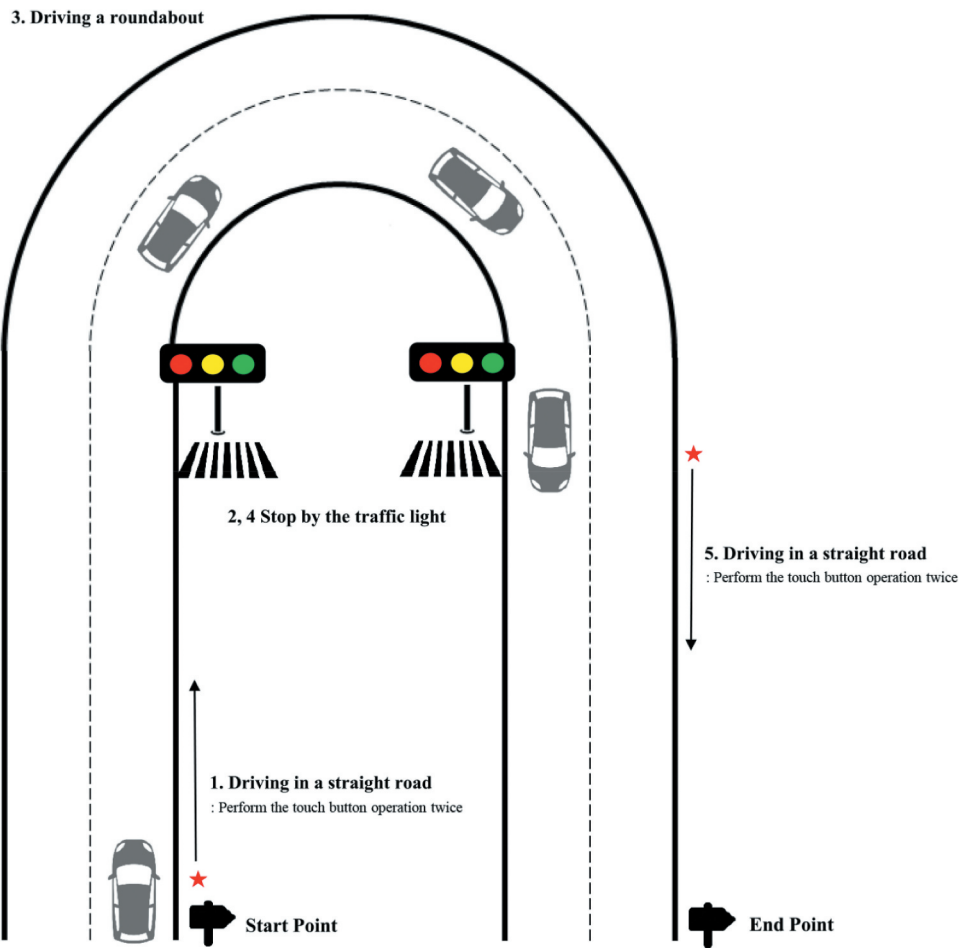


Figure 3. Driving Map and task configuration.

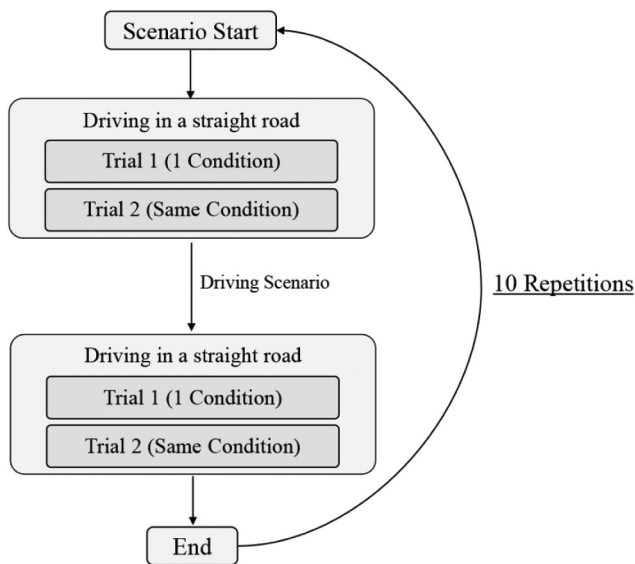


Figure 4. Task process.

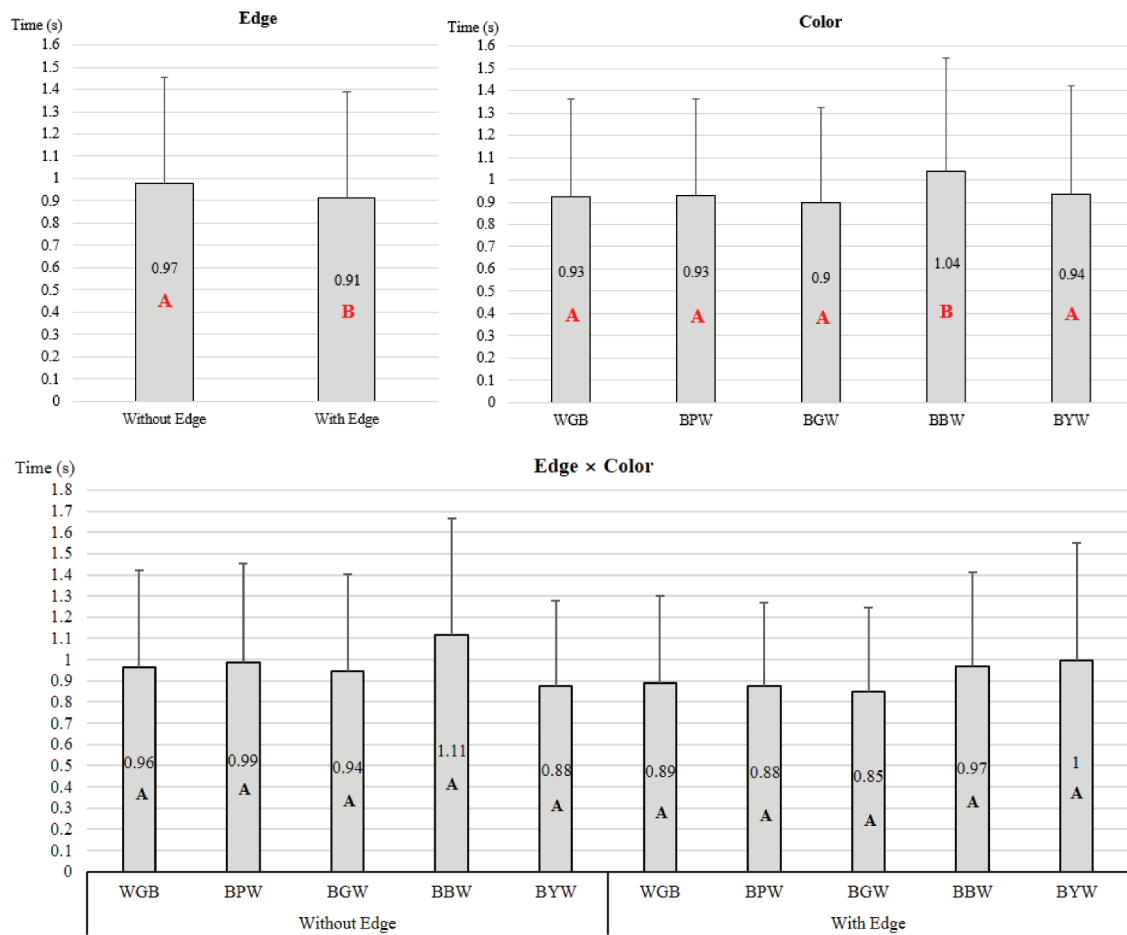
Table 7. P-values for mean completion time and error rate.

	First button presses		All button presses	
	Mean completion time	Error rate	Mean completion time	Error rate
Shape	0.223	0.025a	0.013a	0.561
Edge	0.036a	0.398	0.39	0.498
Color	0.044a	0.581	0.391	0.815
Shape × Edge	0.622	0.815	0.036a	0.554
Shape × Color	0.773	0.564	0.863	0.937
Edge × Color	0.035a	0.976	0.15	0.177
Shape × Color × Edge	0.485	0.436	0.970	0.092

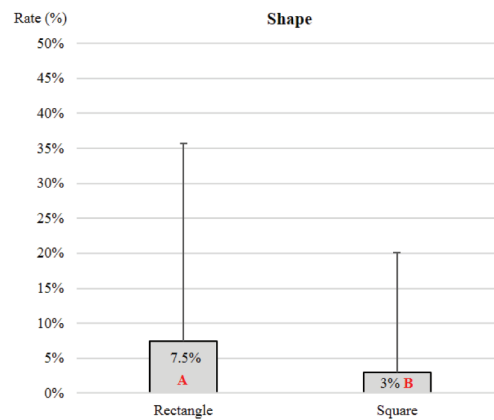
*means significant difference at $p < 0.05$.

(Figure 5(a)). An analysis of the error rate of first button presses showed that there was a statistically significant

effect of Shape. For the rectangle button shape, the error rate was larger than for the square shape (Figure 5(b); $F = 5.942$, p -value < 0.05). An analysis of the mean completion time for all button presses showed that there were statistically significant effects of Shape and Shape × Edge. In the case of the square button shape, the overall mean completion time was shorter (Figure 5(c); $F = 7.575$, p -value < 0.05). A simple effect analysis of Shape × Edge showed that the overall mean completion time for the square button shape was reduced only when there was an



(a) First button presses - mean completion time
(Bars indicated with different letters are significantly different at $p < 0.05$)



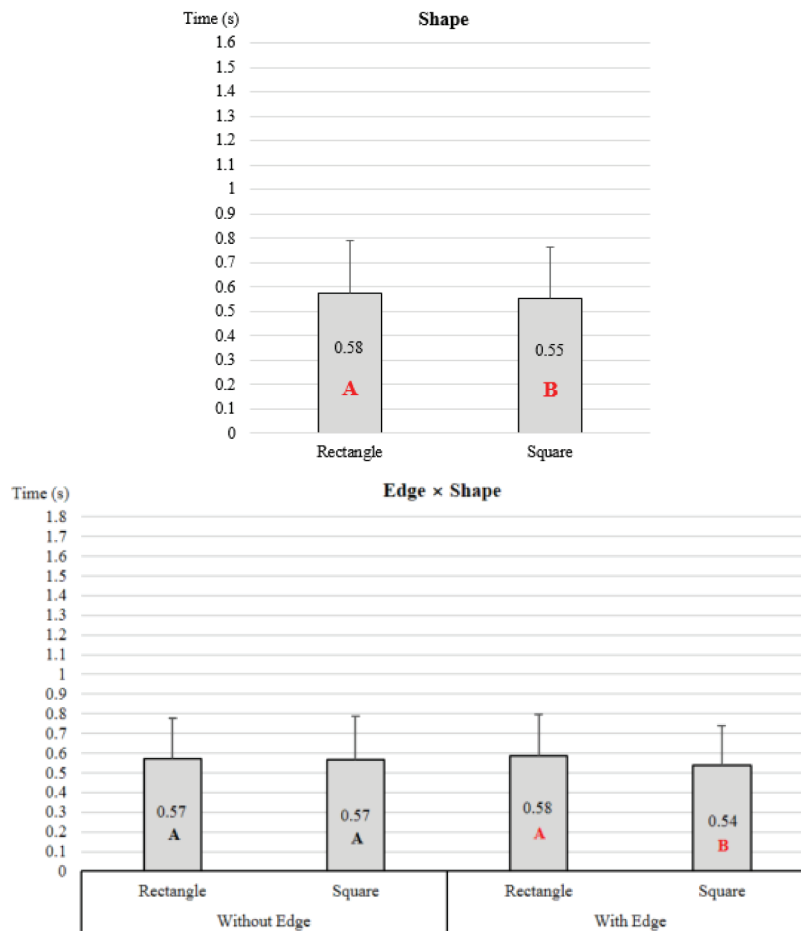
(b) First button presses - error rate
(Bars indicated with different letters are significantly different at $p < 0.05$)

Figure 5. Task Completion Time and Error Rates.

edge (Figure 5(c); $F = 0.036$, p -value < 0.05). There was no significant difference in the error rate for all button presses (Figure 6).

4.2. Subjective preference evaluation

A three-way repeated measures ANOVA was conducted to determine the effects on subjective preferences (six NASA-



(c) All button presses - mean completion time
(Bars indicated with different letters are significantly different at $p < 0.05$)

Figure 5. Continued.

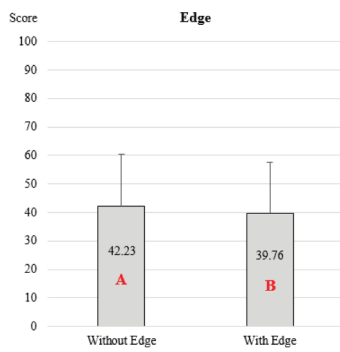
TLX questions, four RCS questions, and four DX questions) such as workload, complexity, and driving experience, of the three independent variables (Shape, Edge, and Color). A post hoc Student Newman-Keuls (SNK) analysis was also conducted. When an interaction was significant, a simple effect analysis was also performed.

4.2.1. NASA-TLX

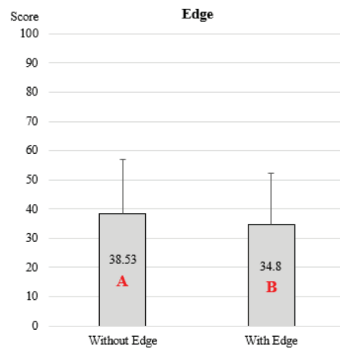
A three-way repeated measures ANOVA was performed on the NASA-TLX question response data ($\alpha = 0.05$) (Table 8). The analysis of mental demand showed that Edge was statistically significant. Mental demand was low when there was a button edge (Figure 6(a); $F = 5.474$, p -value < 0.05) (Figure 5). The analysis results for physical demand, temporal demand, and performance (degree of failure) indicated no statistically significant differences. Effort ($F = 10.364$, p -value < 0.01) and frustration ($F = 5.105$, p -value < 0.05) were significantly lower when there was a button edge, but there were no significant differences for the other factors. (Figure 5(b) and (c)).

4.2.2. RCS

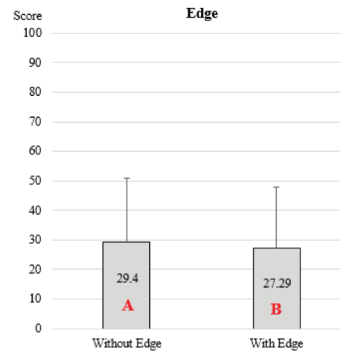
A three-way repeated measures ANOVA was performed on the RCS question response data ($\alpha = 0.05$) (Table 9). The analysis of overall clutter showed that Edge and Edge $\times$ Color were statistically significant. When there was no button edge, overall clutter appeared low ($F = 5.381$, p -value < 0.05). A simple effect analysis of Edge $\times$ Color was conducted, but it was not found to be significant (Figure 7(a)). There were no statistically significant differences in dynamics or inconsistency. For colorfulness, significant differences were detected depending on Shape, Edge, and Color. The colorfulness was high when the button shape was a rectangle and an edge was present. Colorfulness was higher when the color combination was 'background-color: black, button color: blue, font color: white (BBW)' or 'background-color: black, button color: yellow, font color: white (BYW)', than when it was 'background-color: white, button color: gray, font color: black (WGB)', 'background-color: black, button color: purple, font color: white (BPW)', or 'background-color: black, button color: gray, font color: white (BGW)' (Figure 7(b)).



(a) Mental demand

(Bars indicated with different letters are significantly different at $p < 0.05$)

(b) Effort

(Bars indicated with different letters are significantly different at $p < 0.05$)

(c) Frustration

(Bars indicated with different letters are significantly different at $p < 0.05$)**Figure 6.** NASA-TLX results.

(Color order: Background-Button-Font; Abbreviation: WGB = White-Gray-Black; BPW = Black-Purple-White; BGW = Black-Gray-White; BBW = Black-Blue-White; BYW = Black-Yellow Green-White)

Table 8. P-values for NASA-TLX.

	NASA-TLX					
	1. Mental Demand	2. Physical Demand	3. Temporal Demand	4. Effort	5. Performance(degree of failure)	6. Frustration
Shape	0.273	0.668	0.712	0.838	0.481	0.654
Edge	0.03a	0.215	0.702	0.005aa	0.386	0.036a
Color	0.475	0.698	0.714	0.331	0.937	0.558
Shape $\times$ Edge	0.443	0.552	0.976	0.779	0.18	0.706
Shape $\times$ Color	0.104	0.768	0.8	0.91	0.706	0.129
Edge $\times$ Color	0.076	0.11	0.126	0.78	0.474	0.164
Shape $\times$ Color $\times$ Edge	0.207	0.382	0.175	0.828	0.423	0.732

*means significant difference at $p < 0.05$. **means significant difference at $p < 0.01$

Table 9. P-values for RCS.

	RCS (Reduced Clutter Score)			
	1. Overall Clutter	2. Dynamics	3. Inconsistency	4. Colorfulness
Shape	0.381	0.915	0.277	0.048a
Edge	0.032a	0.368	0.506	0.000aa
Color	0.075	0.713	0.886	0.044a
Shape $\times$ Edge	0.37	0.725	0.736	0.428
Shape $\times$ Color	0.323	0.972	0.279	0.191
Edge $\times$ Color	0.009aa	0.196	0.066	0.076
Shape $\times$ Color $\times$ Edge	0.846	0.056	0.718	0.985

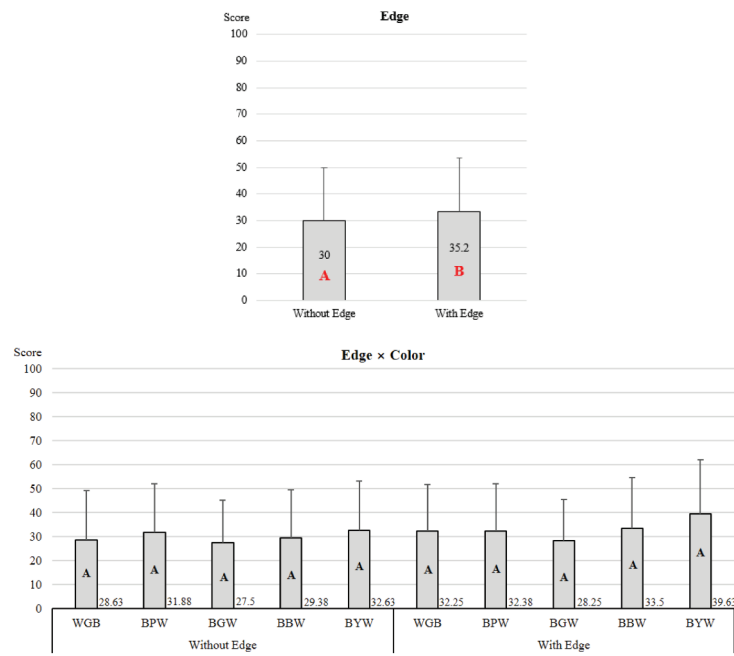
*means significant difference at $p < 0.05$. **means significant difference at $p < 0.01$.

4.2.3. Driving user experience (DX)

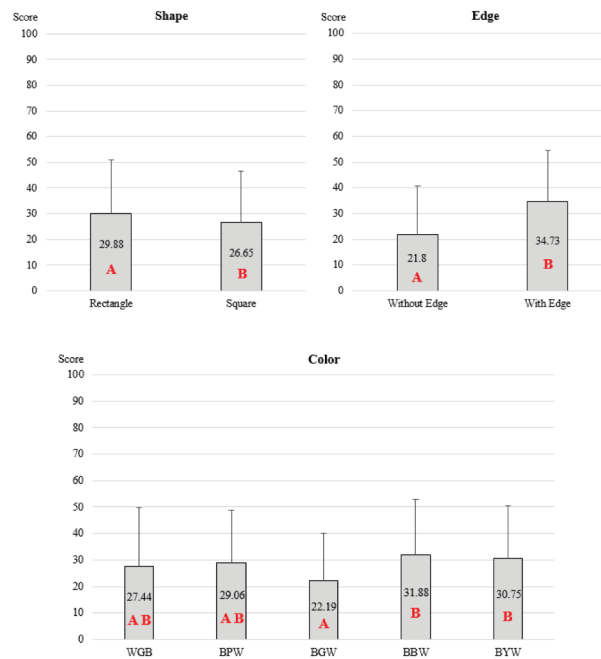
A three-way repeated measures ANOVA was performed on DX ($\alpha = 0.05$) (Table 10). The analysis of familiarity indicated that only Edge $\times$ Color had a significant effect (Figure 8(a)). A simple effect analysis was performed, but it was not found to be statistically significant. The analysis of learnability indicated that it was significantly affected only by Shape $\times$ Color $\times$ Edge (Figure 8(b)). However, a simple effect analysis indicated that it was not statistically significant. The analysis of intuition showed that only Edge $\times$ Color had statistically significant effect (Figure 8(c)). A simple effect analysis, however, showed that it was not statistically significant. The analysis of overall usability indicated that there was a statistically significant effect of Edge. When there was an edge, overall usability was high (Figure 8(d)).

4.3. Eye glances

A three-way repeated measures ANOVA was performed for mean glance time and number of glances for on-road and off-road conditions ($\alpha = 0.05$) (Table 11). Analyzing the mean glance time and number of glances for on-road glances detected no statistically significant effect. The analysis of mean glance time for off-road glances detected a significant effect only for Shape $\times$ Color. A simple effect analysis of Shape $\times$ Color showed that when the color combination was BBW, the square button shape had a longer mean glance time than the rectangle button shape (Figure 9; $F = 3.9238$, p -value < 0.05). No factors or interactions had statistically significant effects on the number of off-road glances (Figure 9).



(a) Overall clutter
(Bars indicated with different letters are significantly different at $p < 0.05$)



(b) Colorfulness
(Bars indicated with different letters are significantly different at $p < 0.05$; In case of Edge $p < 0.01$)

Figure 7. RCS results.

(Color order: Background-Button-Font; Abbreviation: WGB = White-Gray-Black; BPW = Black-Purple-White; BGW = Black-Gray-White; BBW = Black-Blue-White; BYW = Black-Yellow Green-White)

5. Discussion

5.1. Interface elements

To provide guidance on vehicle interfaces, recommendations concerning the main effects of each dependent variable are presented in Table 12. For the time index and error rate, Times, NASA-TLX, and RCS, variables with low scores are

recommended. For DX, a variable with a high score is recommended.

5.1.1. Effect of touch button shape

The results for the overall quantitative indicators indicate that the square button shape resulted in better task performance (shorter mean completion time and lower error

Table 10. P-values for DX.

	DX (Driving User Experience)			
	1. Familiarity	2. Learnability	3. Intuition	4. Overall Usability
Shape	0.667	0.403	0.121	0.338
Edge	0.521	0.053	0.081	0.028a
Color	0.319	0.734	0.384	0.522
Shape × Edge	0.642	0.782	0.669	0.433
Shape × Color	0.216	0.862	0.512	0.104
Edge × Color	0.027a	0.243	0.019a	0.133
Shape × Color × Edge	0.256	0.014a	0.636	0.916

*means significant difference at $p < 0.05$.

rate) than the rectangle shape (Table 12). Previous research has shown that button operation performance varies depending on the size of the fingertip of the operator and the ratio of fingertip size to the touch area (Jung & Im, 2015; Tao et al., 2018). In the case of a touch button, because the size of the real touchable area is smaller than the recognized size (Jung & Im, 2015), the proportion of the area should be considered more important than in the case of a physical keypad. That is why Suh and Ferris (2019) showed better task performance with a physical keyboard than with touch buttons. In this study, we compared square and rectangular keypads using the same diagonal length. Through experiments, we conclude that the square condition was excellent, but it is difficult to ascertain whether the shape of the button itself is important. However, based on the experimental results, it is recommended that the vertical length be not too short relative to the horizontal length.

A critical issue exists in the guidelines for selecting the sizes of touch buttons. None of the IVIS studies listed in Table 1 considered the driver's visual angle for the target button. Even buttons of the same size may feel larger or smaller depending on the distances from the driver's eye to where on the screen the buttons are displayed. The distances would vary depending on the vehicles used in the previous studies, the type of driving simulator used, and the eye levels of the participants. For this reason, it is judged that there are differences in the recommendations obtained from previous IVIS touch button studies (Table 1) that make them difficult to utilize directly. When selecting a button size, it is necessary to employ a button standard through visual angle calculation using the recommended value of 86 arcmin and the minimum value of 41 arcmin established by the NHTSA (Campbell et al., 2007).

5.1.2. Effect of touch button edge

The analysis of overall quantitative indicators revealed that task performance (short execution time and low error rate) was high when there was an edge (Table 12). Studies have shown that touch buttons pose a threat to driving safety because the use of touch buttons is more dependent on vision than the use of physical buttons that offer nonvisual cues to senses such as touch and hearing (Fitch et al., 2013; Suh & Ferris, 2019). The presence of an edge helps a driver to perceive the touchable area of the touch button. As a result, the interaction between the driver and the touch button becomes clear, and the overall

task performance and usability appear to be good. On the other hand, distraction appears to be high when there is an edge because the presence of the edge increased the complexity of the screen. The orange (VIVID) color used as the edge color in this study has a very low contrast ratio with most colors (Table 4) and is a color commonly used in actual vehicles. Therefore, it seems to function more favorably in terms of RCS than other colors. In addition, the overall performance, workload, and usability results indicate that there are few negative effects of the orange (VIVID) color edge. Therefore, for the driver's safety, it is desirable to use an edge that can present a clear border of the touch button and an edge that has a relatively low contrast rate.

5.1.3. Effect of color combination

Color combination recommendations are made to avoid the adverse effects of BBW and BYW on the completion time (first button press) and colorfulness (RCS). The color combination greatly affects the legibility of the system information display (Humar & Gradis, 2008). When the legibility of information is low, the reading time increases (Naujoks et al., 2019). As a result, the task efficiency of the driver is reduced, and this may threaten driving safety. The contrast ratio can be used as a quantitative indicator of legibility (Naujoks et al., 2019; Stevens et al., 2002). Naujoks et al. (2019) suggest that the contrast ratio of information display colors should be at least 3:1. For the BBW color combination, the contrast ratio when the blue color (VIVID) is used as the button color and the black color (VIVID) is used as the background color was as low as 2.89:1 (Table 4). Humar and Gradis (2008) quantitatively assessed legibility using the average value of characters correctly identified among the characters shown to participants in a study comparing 56 color combinations for font and background color. The combination of blue and black had very low legibility, ranked 50th out of 56. The BBW combination used in this study was judged to have a high mean completion time (first button presses) and RCS (colorfulness) because of the low legibility of the background color and button color. However, even if an appropriate contrast ratio is used, legibility may change due to other factors (Sanders & McCormick, 1998). In terms of contrast ratio, the use of the BBW combination should be avoided, but if a BBW combination is used, the use of a rectangle button shape is recommended, in accordance with the significant interaction effect detected in this study (Table 12). In addition, considering the low contrast between the background and the button, it is recommended to use an edge. The yellow-green color (VIVID) used as the button color in the BYW combination has a very low contrast ratio of 2.08:1 when white (VIVID) is used as the font color (Table 4). Humar and Gradis (2008) did not examine the contrast ratio of the yellow-green/white combination, but they did examine the combinations of yellow/white and green/white and found them to rank 56th of 56 and 42nd of 56, respectively, in legibility. Therefore, it is judged that the mixed index yellow-green also has low legibility and high RCS (colorfulness). If yellow-green is used as the button color, the contrast with the font color needs to be changed. In the case of the

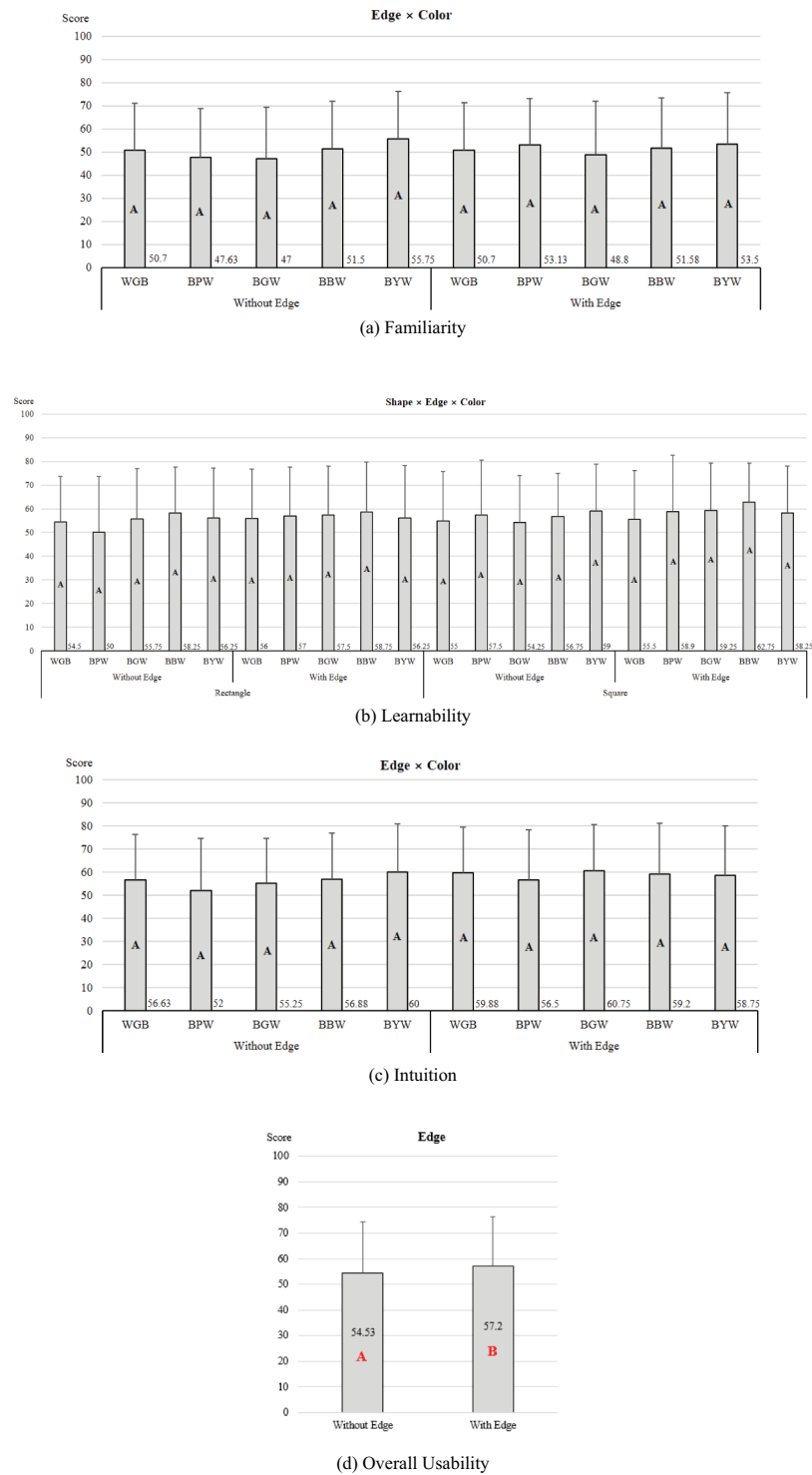


Figure 8. DX results.

(Color order: Background-Button-Font; Abbreviation: WGB = White-Gray-Black; BPW = Black-Purple-White; BGW = Black-Gray-White; BBW = Black-Blue-White; BYW = Black-Yellow Green-White)

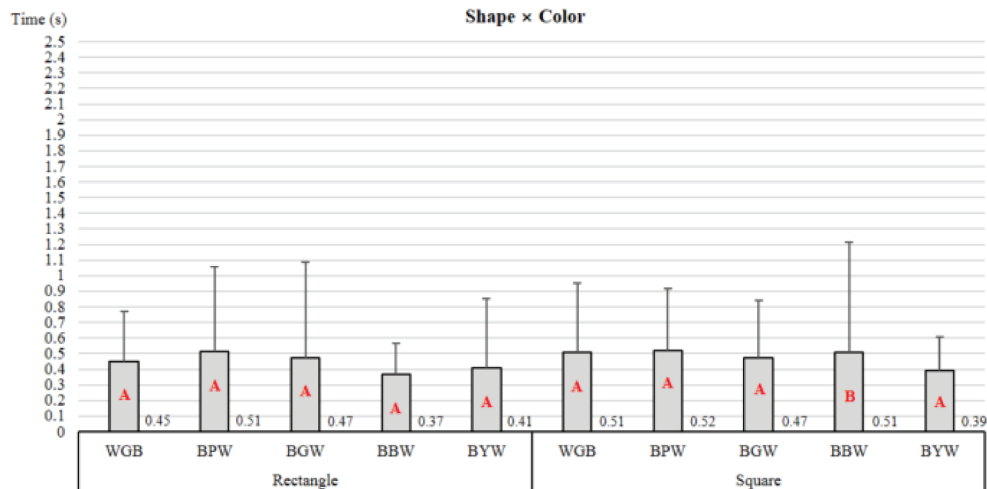
BPW color combination, the contrast ratio when black (VIVID) is used as the background color and purple (VIVID) is used as the button color is very low, at 1.48:1 (Table 4). However, the contrast ratio with purple (VIVID) used as the button color and white (VIVID) used as the font color is 5.96:1, which indicates very good contrast even at night (Table 4; Naujoks et al., 2019). As a result, this combination appears to have good overall usability. However, it is

recommended to use an edge, considering the contrast between the background and the buttons. The BGW and WGB color combinations are composed only of achromatic colors, without color tone. Therefore, although the contrast ratio of gray (VIVID) to white (VIVID) is 2.58:1, it appears that the resulting overall usability is good because of the low RCS (colorfulness) (Table 4; Table 12). However, in the case of the WGB color combination, it is recommended to use an

Table 11. P-values for glance data.

	On-Road Glance		Off-Road Glance	
	Mean glance time	Number of glances	Mean glance time	Number of glances
Shape	0.172	0.200	0.33	0.877
Edge	0.231	0.571	0.765	0.980
Color	0.133	0.648	0.1035	0.921
Shape × Edge	0.809	0.974	0.3854	0.905
Shape × Color	0.820	0.598	0.006aa	0.053
Edge × Color	0.537	0.122	0.857	0.962
Shape × Color × Edge	0.606	0.598	0.564	0.453

*means significant difference at $p < 0.05$.



(a) Mean glance time (off-road)

(Bars indicated with different letters are significantly different at $p < 0.05$)

Figure 9. Eye glance results.

(Color order: Background-Button-Font; Abbreviation: WGB = White-Gray-Black; BPW = Black-Purple-White; BGW = Black-Gray-White; BBW = Black-Blue-White; BYW = Black-Yellow Green-White)

Table 12. Summary of recommendations for Shape, Edge, Color, and interactions.

Data classification	Indicator	Recommended values (for statistically significant effects)			
		Shape	Edge	Color	Interaction
Task performance (First button presses)	Mean completion time (s)	-	with edge	Except BBW	-
Task performance (All button presses)	Error rate (%)	square	-	-	-
NASA-TLX (0–100 Score)	Mean completion time (s)	square	-	-	with Edge: square
	Error rate (%)	-	-	-	-
	Mental demand	-	with edge	-	-
	Physical demand	-	-	-	-
	Temporal demand	-	-	-	-
	Effort	-	with edge	-	-
	Performance (degree of failure)	-	-	-	-
	Frustration	-	with edge	-	-
RCS (0–100 Score)	Overall clutter	-	without edge	-	-
	Dynamics	-	-	-	-
	Inconsistency	-	-	-	-
	Colorfulness	square	without edge	BGW, WGB, BPW	-
DX (0–100 Score)	Familiarity	-	-	-	-
	Learnability	-	-	-	-
	Intuition	-	-	-	-
	Overall usability	-	with edge	-	-
Eye tracking data	On-road mean glance time (s)	-	-	-	-
	Number of on-road glances	-	-	-	-
	Off-road mean glance time (s)	-	-	-	-
	Number of off-road glances	-	-	-	BBW: rectangle

edge, considering the contrast between the background and buttons.

5.2. First button presses

When attention is divided by separate tasks in the workspace, overall task performance decreases (Fitch et al., 2015; Lyngsie et al., 2013; Tippey et al., 2017). This makes it difficult to explore and accomplish the goals of the work. In the case of driving, attention is divided between driving, which is the main task, and IVIS, which is an auxiliary task (Li, Chen et al., 2017). IVIS work increases a driver's cognitive load and reduces performance at driving tasks (Fitch et al., 2015; Lyngsie et al., 2013; Tippey et al., 2017). Therefore, the majority of IVIS studies have measured task completion time (Angelini et al., 2016; Jeong & Liu, 2017; Lee et al., 2019; Weidner & Broll, 2018). However, the demands associated with visually navigating a target in a button press operation are greatest when the first button is pressed. It is for this reason that, in this study, the average time taken to press the first button was greater than the average times taken for the remaining button operations (Table 13). Therefore, it is necessary to consider analyzing task performance with more granularity in future studies.

There were statistically significant differences with respect to Edge and Color for the mean completion time for first button presses but no differences in mean completion time for 'all button presses'. This means that Edge and Color affect the time it takes the driver to navigate the keypad for the first time. In the presence of an edge, it is judged that legibility increases because of a clear perception of the touchable area, resulting in a decrease in information reading time (Figure 3; Naujoks et al., 2019). In addition, in the case of the BBW color combination, the contrast ratio is low, legibility is lowered, and

information reading time is increased (Figure 3; Naujoks et al., 2019). The learnability results for the DX scale indicate no significant differences with respect to Edge or Color. This means that there is no difference in the level of learnability based on presence or absence of an edge or the color combination. Therefore, no significant difference is judged to exist for the mean completion time for 'all button presses'.

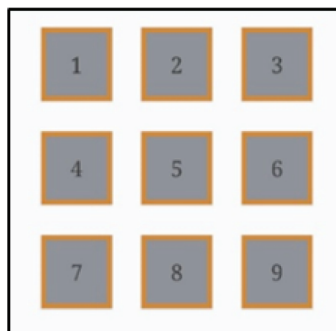
5.3. Limitations and future research

This study was limited in the number of touch button shapes, color combinations, and edge conditions considered in the interface design. Only two button shapes, rectangle and square, were used in the experiment. No significant differences were detected with respect to button shape. Further research on various shapes and symbols is needed. In this study, color combinations for the background, buttons, and fonts were selected based on suggestions in the literature and the results of research on actual cars. While the number of possible combinations is very large, only some color combinations were included in this study to avoid oversizing the experiment. Further research is needed to study more color combinations. In this study, when an edge was present, orange was used for the edge, given that white was used for the buttons and fonts, and the contrast with other colors was low. For this reason, the presence of an edge may influence the effect of the color of the touch button. In the future, it is necessary to conduct detailed studies in which various edge colors and thicknesses are considered.

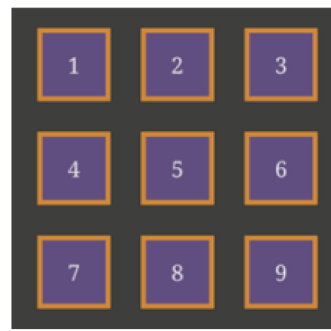
Many vehicles have touch screens (Suh & Ferris, 2019), but the majority are limited to the infotainment system. On the other hand, the number of physical buttons on steering wheels is increasing because of the addition of functions

Table 13. Mean completion time comparison by button presses.

	Firstbutton presses	Second button presses	Thirdbutton presses	Fourth button presses	Fifthbutton presses
Mean Completion Time (s)	0.947	0.47	0.47	0.46	0.47



(Square, With Edge, WGB)



(Square, With Edge, BPW)

Figure 10. Two recommended keypad combinations.

(Edwards et al., 2017). The results of this IVIS study, which was focused on the operation of touch buttons, can be used as basic information for the design of the touch button working area, but the results are not well suited for use as basic information for the design of physical keypads in vehicles. In this study, considering that the difficulty of touch button tasks is affected by a number of conditions, a relatively easy touch button task (five-digit number input) was selected. Accordingly, familiarity, learnability, and mean completion time for 'all button presses' may not be significant.

The interpretation of the results of this study may seem too trivial from the perspective of some regulatory agencies or standards. For example, a rectangular button appears to take longer statistically than a square button, but the time it takes to press a rectangular button is already small, indicating that this is not a problem. However, this may be because the task in this experiment is too simple. If a sufficiently complex task is carried out, as in a real driving situation, it may cause safety problems. From the perspective of the current standard, we expect that even if there are no problems, the data from this study can be helpful in improving driving safety and usefulness for further research.

6. Conclusions

In this study, we investigated the effect of touch-button interface factors (touch button shape, color combination, and edge presence or absence) on IVIS usability. An objective task performance measurement was conducted using a driving simulator and eye tracker, and a subjective preference assessment was conducted using previously developed evaluation tools (NASA-TLX, RCS, and DX). For button keypads having the same diagonal length, a square button keypad generally has better performance and usability than a rectangular button keypad. This indicates that touch button task performance is related to the proportion of the fingertip size to the area of the touch button. The presence of an edge generally resulted in better performance and usability than the absence of an edge. This is because the edge clearly suggests to the driver the area of the touch button and makes button recognition easier, improving touch button task performance. The BBW and BYW color combinations, with low contrast ratios, had low legibility, and their overall performance and usability were poor. The BBW requires the use of an edge because of the effect of the contrast between the button and background colors, and the BYW combination requires a change in the font color. WGB and BPW, which were generally usable color combinations, require the use of an edge because of the contrast between the button and background colors. Therefore, in this study, some recommended combinations of WGB and BPW using a square shape and border are presented (Figure 10). In this study, independent touch button variables were defined using objective indicators (visual angles, contrast ratio table (VIVID), and a vehicle survey). In addition, differences in performance and preference for independent variable conditions were compared and analyzed by various evaluation tools. The results of this study have yielded recommendations for touch button shape, edge presence or absence, and color combination

for effective information display and input, and the findings can be used as a basis for further research.

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